

Engineering Infrastructure & System Architecture Blueprint

CLASSIFICATION / ACCESS LEVEL: ENGINEERING & TECHNICAL OPERATIONS

This technical blueprint describes the multi-region microservices infrastructure supporting the DMS Solutions Core Platform.

Cloud Infrastructure: All production workloads run on Amazon Web Services (AWS) across two primary regions: us-east-1 (Primary) and us-west-2 (Disaster Recovery Failover). Compute is managed via Amazon Elastic Kubernetes Service (EKS).

Service Topology & Event Streaming: Our backend consists of 45 containerized Go and Python microservices communicating asynchronously via an Apache Kafka distributed event bus with a 7-day retention period. Synchronous inter-service calls use gRPC with Protobuf schemas.

Databases & Caching Layer: Transactional application data is persisted in Amazon Aurora PostgreSQL with multi-AZ replication. Redis Enterprise clusters provide sub-millisecond in-memory caching and distributed session management.

High Availability & SLA: The infrastructure is designed for 99.99% uptime with automated health checks, multi-zone pod autoscaling, and cross-region database read-replicas.